

Equipment Inspection Report #07 — Boiler Feedwater Pump B-102

Date: August 24, 2026 | **Inspector:** R. Sharma, Senior Diagnostics Specialist

Facility: SRM/SIH Thermal Power Unit — Zone 4 High Pressure Loop

1. Operational Baseline & Telemetry Data

Boiler Feedwater Pump B-102 underwent routine vibration analysis and thermal imaging following scheduled operational hour threshold (4,500 hours).

Vibration Analysis Findings:

Vibration level measured at **7.8 mm/s RMS** on drive-end bearing housing (B-102-BRG). This significantly exceeds the ISO 10816 Class II alert threshold of **4.5 mm/s RMS** and mandatory action threshold of 7.0 mm/s RMS.

Thermal Imaging & Cavitation Observations:

Thermal camera spectrum recorded temperature of 88.4°C on drive-end sleeve casing. Acoustic emissions detect clear cavitation spikes within impeller stage 2 due to restricted inlet flow at Suction Strainer ST-01.

2. Criticality Assessment & Action Plan

CRITICALITY RATING: HIGH (Category 2 Alert). Continued operation poses severe risk of catastrophic bearing seizure and impeller destruction. Mandatory maintenance clearance required immediately.